

6.006: Introduction to Algorithms

Year: Spring 2020

Instructor(s): Prof. Erik Demaine, Dr. Jason Ku, Prof. Justin Solomon

OCW URL: <https://ocw.mit.edu/courses/6-006-introduction-to-algorithms-spring-2020/>

Textbooks

[Optional] CLRS - Introduction to Algorithms, 3rd ed, MIT Press

Available Resources

- 20 Lecture Notes (PDF)
- 20 Video Lectures
- 20 Recitation Notes
- 8 Problem Sets (with solutions)
- 3 Quizzes + Final Exam
- 9 Problem Sessions (with solutions)

#	Topic	Readings	Resources
PART I: Foundations & Sorting			
Wk 1	Lec 1: Introduction	Rec 1-2 / ProbSess 1	Notes x2 / Video x2 / PS0
Wk 2	Lec 3: Sorting	Rec 3-4 / ProbSess 2	Notes x2 / Video x2 / PS1
Wk 3	Lec 5: Linear Sorting	Rec 5 / ProbSess 3	Notes / Video / PS2
PART II: Trees & Heaps			
Wk 4	Lec 6: Binary Trees, Part 1	Rec 6-7 / ProbSess 4	Notes x2 / Video x2 / PS3
Wk 5	Lec 8: Binary Heaps	Rec 8-9 / Quiz 1 Rev	Notes x2 / Video x2 / PS4
PART III: Graph Algorithms			
Wk 6	Lec 10: Depth-First Search	Rec 10-11 / ProbSess 5	Notes x2 / Video x2
Wk 7	Lec 12: Bellman-Ford	Rec 12 / ProbSess 6	Notes / Video / PS5 / Quiz 1
Wk 8	Lec 13: Dijkstra's Algorithm	Rec 13-14 / ProbSess 7	Notes x2 / Video x2 / PS6 / Quiz 2 Rev
PART IV: Dynamic Programming			
Wk 9	Lec 15: DP Part 1: SRTBOT, Fib, DAGs	Rec 15-16 / ProbSess 8	Notes x2 / Video x2
Wk 10	Lec 17: DP Part 3	Rec 17	Notes / Video / PS7 / Quiz 2
Wk 11	Lec 18: DP Part 4: Pseudopolynomial	Rec 18-19 / ProbSess 9	Notes x2 / Video x2 / PS8 / Quiz 3 Rev
PART V: Wrap-up			
Wk 12	Lec 20: Course Review	Rec 20	Notes / Video / Quiz 3
Wk 13	Lec 21: Algorithms - Next Steps	--	Video only
Wk 14	Final Exam	--	Final Exam (3 hr)